Name_____ Date_____ Period_____

LESSON 5-5 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Volume of Rectangular Prisms

Maryland Standard 6.G.A.2 · Learning target – I can find the volume of a rectangular prism by multiplying length, width, and height.

REMEMBER

- **Volume** counts the unit cubes that fill a solid. Use **cubic units** (cm³, in³, ft³).
- **V = l × w × h**, or **V = B × h**, where *B* is the area of the base.

WORKED EXAMPLE

A box is 7 cm long, 3 cm wide, and 5 cm tall.

Step 1 One layer of cubes: B = 7 × 3 = 21

Step 2 Stack 5 layers: 21 × 5 = 105

Answer **105 cm³**

WORD BANK

volume – the space inside a solid

cubic units – cm³, in³, ft³

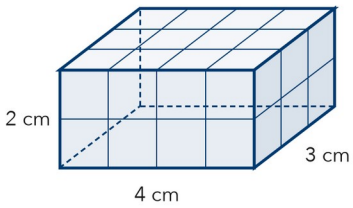
base area – length × width of the bottom

layer – one flat level of cubes

PRACTICE

show your work · label units

1. Each small cube is 1 cm³. Find the volume of the prism.



Step 1 Count the cubes in one layer: 4 × 3.

Step 2 Count the layers.

Step 3 Multiply.

Show your work.

Answer_____ cm³

2. Find the volume of each rectangular prism.

a. 8 in × 5 in × 2 in _____ in³

b. 10 cm × 10 cm × 4 cm _____ cm³

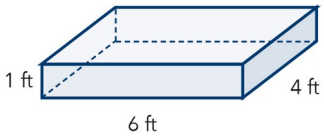
c. 6 ft × 6 ft × 6 ft _____ ft³

d. B = 45 m², h = 7 m _____ m³

e. 12 m × 3 m × 5 m _____ m³

f. B = 32 in², h = 9 in _____ in³

3. Sandbox. A sandbox is 6 ft long, 4 ft wide, and 1 ft deep. One bag of sand fills 2 ft³. How many **bags** fill the sandbox?



Step 1 Find the volume: 6 × 4 × 1.

Step 2 Divide by 2 ft³ for each bag.

Show your work.

Answer_____ bags

4. Fish Tank. A fish tank is 20 in long and 10 in wide. It holds 3,000 in³ of water when full. How tall is the tank?

Step 1 Find the base: 20 × 10.

Step 2 Divide 3,000 by the base.

Answer_____ in

5. Compare. Box A is 6 in × 4 in × 2 in. Box B is 3 in × 4 in × 4 in. Mika says Box A holds more because it is longer. Is she right? Explain.

Frame *Mika is ___ because Box A holds ___ in³ and Box B holds ___ in³.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Find a box at home. Measure or estimate its length, width, and height, and find its volume.

Name_____ Date_____ Period_____

LESSON 5-5 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Missing Fractional Edges

Maryland Standard 6.G.A.2 · Learning target – I can find a missing edge of a prism and find volumes with fractional edge lengths.

REMEMBER

- **V = l × w × h** works with fractions too. Write mixed numbers as fractions first.
- Missing edge: **divide** the volume by the product of the other two edges.

WORKED EXAMPLE

A box is 2 ½ ft × 2 ft × 1 ½ ft. Find its volume.

Step 1 Write as fractions: 5/2 × 2/1 × 3/2

Step 2 Multiply: (5 × 2 × 3)/(2 × 1 × 2) = 30/4

Step 3 Simplify: 30/4 = 7 ½

Answer **7 ½ ft³**

WORD BANK

fraction edge – an edge like 1/2 ft or 1 1/2 in

missing edge – the edge you do not know

divide – ÷ undoes ×

product – the answer when you multiply

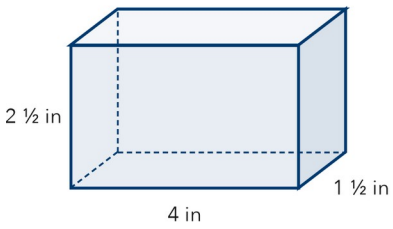
PRACTICE

show your work · label units

1. Find the volume of the prism.

Step 1 Write 1 ½ as 3/2 and 2 ½ as 5/2.

Step 2 Multiply 4 × 3/2 × 5/2.



Show your work.

Answer_____ in³

2. Find each volume.

a. ½ ft × 6 ft × 8 ft _____ ft³

b. 3 ½ m × 2 m × 4 m _____ m³

c. 2 ½ cm × 4 cm × 3 cm _____ cm³

d. 1 ½ in × 1 ½ in × 4 in _____ in³

e. 4 ft × ½ ft × 5 ft _____ ft³

f. 6 cm × 1 ½ cm × 3 cm _____ cm³

3. Planter. A planter box is 3 ft long and 1 ½ ft wide. It holds 9 ft³ of soil. How **deep** is it?

Step 1 Multiply the two edges: 3 × 1 ½.

Step 2 Divide 9 by that answer.

Answer_____ ft

4. Small Cubes. A box is 2 in long, 1 in wide, and 1 ½ in tall. It is packed with cubes that are ½ in on each edge.

Step 1 Count the cubes along each edge: 2 ÷ ½, 1 ÷ ½, 1 ½ ÷ ½.

Step 2 Multiply to get the number of cubes.

a. How many cubes fill the box? _____

b. Each cube is 1/8 in³. What is the volume of the box? _____ in³

TEST PRACTICE

choose the best answer

5. A box is 5 in long, 2 in wide, and 1 ½ in tall. What is its volume?

A 8 ½ in³ **B** 10 in³ **C** 15 in² **D** 15 in³

6. A prism has a volume of 96 ft³. Its base is 4 ft by 6 ft. What is its height?

Step 1 Find the base: 4 × 6.

Step 2 Divide 96 by the base.

A 4 ft **B** 16 ft **C** 24 ft **D** 576 ft

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Find a box at home with a fraction edge (like 2 ½ in). Estimate its volume.